

Nested If Assignment Problems. Do the IPO and code for each of the problems below.

- 1) The student will enter their last name and score. Determine their letter grade using the scale below. Display the student last name and letter grade.

Score	Letter Grade
90 & up	A
80 to 89	B
70 to 79	C
60 to 69	D
Below 60	F

Input	Process	Output
• Student's Last Name • Score	• Assign a variable to the input for Student's Last Name • Assign a variable to the input for Score • Declare Letter Grade and set as A if Score is greater than or equal to 90% ○ If Score is greater than or equal to 80%, set Letter Grade as B ○ If Score is greater than or equal to 70%, set Letter Grade as C ○ If Score is greater than or equal to 60%, set Letter Grade as D ○ Otherwise, set Letter Grade as F • Display Student's Last Name and Letter Grade	• Student's Last Name • Letter Grade

- 2) You are buying apples in bulk. Enter the quantity in pounds, determine the price per pound, then display the price per pound and total.

LBS Price Per Pound

>100 .10

50-100 .25

Under 50 .50

Input	Process	Output
Quantity (lbs.)	- Assign a variable to the input for Quantity - Declare Price/lbs. and set as \$0.10 if Quantity is greater than 100 - If Quantity is greater than or equal to 50, set Price/lbs. as \$0.25 - Otherwise, set Price/lbs. as \$0.50 - Calculate Total by finding product of Quantity and Price/lbs. - Display Price/lbs. and Total	- Price/lbs. - Total

- 3) Enter the employee last name, hours worked and job code. Compute the pay based on the hourly rate per the job code. Display employee last name, hours worked, pay rate and total.

Job Code	Pay Rate
E	25.00
J	20.00
A	15.00

Input	Process	Output
• Employee's Last Name • Hours Worked • Job Code	• Assign variables to the input for Employee's Last Name, Hours, Job Code • Declare Pay Rate and set as \$25.00 if Job Code equals E ○ If Job Code equals J, set Pay Rate as \$20.00 ○ If Job Code equals A, set Pay Rate as \$15.00 • Calculate Total by finding the product of Hours and Pay Rate • Display Employee's Last Name, Hours Worked, Pay Rate, Total	• Employee's Last Name • Hours Worked • Pay Rate • Total

- 4) Allow the user to enter the annual salary. Determine the tax rate from the table below. Compute the tax amount owed. Display salary, tax rate and tax amount.

Salary	Tax Rate
>100,000	40%
50,000 - 100,000	35%
Under 50,000	25%

Input	Process	Output
• User's Annual Salary	• Assign a variable to the input for Annual Salary • Declare Tax Rate and set as 40% (.40) if Annual Salary is greater than 100,000 ○ If Annual Salary is greater than or equal to 50,000, set Tax Rate as 35% (.35) ○ Otherwise, set Tax Rate as 25% (.25) • Calculate Tax Amount by finding the product Annual Salary and Tax Rate • Display User's Annual Salary, Tax Rate, and Tax Amount	• User's Annual Salary • Tax Rate • Tax Amount

- 5) You are running a metal recycling center and must pay people for metals they bring in. You give them a rate based on the weight in the table below. Allow the user to enter the weight. Determine the rate and then display the weight, rate and total given to the customer.

Weight	Rate Per Pound
>100	.50
30-100	.25
20- less 30	.20
Less 20	.10

Input	Process	Output
• Weight	• Assign a variable to the input for Weight • Declare Rate/lbs. and set as \$0.50 if Weight is greater than 100 ○ If Weight is greater than or equal to 30, set Rate/lbs. as \$0.25 ○ If Weight is greater than or equal to 20, set Rate/lbs. as \$0.20 ○ Otherwise, set Rate/lbs. as \$0.10 • Calculate Total by finding the product of Weight and Rate/lbs. • Display Weight, Rate/lbs., and Total	• Weight • Rate/lbs. • Total